Reproducing an Explainable Financial News Sentiment Toolkit: Baselines, Fine-Tuned FinBERT, and Robustness Analysis

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Abstract

This work reproduces and extends the *xai-finnews-sentiment* toolkit presented in this article (Cristescu and Mara, 2025), this is an explainable financial news sentiment system that applies multiple NLP approaches: Loughran–McDonald (LM) lexicon, VADER, TF-IDF + Logistic Regression (LR), and domain-adapted FinBERT, to financial news from Financial PhraseBank (FPB), FiQA-2018 headlines, and a manually labelled Marketaux corpus. The primary evaluation metric is macro-averaged F1, chosen to account for class imbalance across the three sentiment classes (negative, neutral, positive). The key result is successfully reproduced: fine-tuning FinBERT (Araci, 2019) on gold annotations achieves a macro-F1 of **0.766** on the held-out test set, outperforming the best lexicon baseline (LM: 0.552) and zero-shot FinBERT (0.645) by a substantial margin. A reduced-data ablation reveals that the TF-IDF+LR baseline retains 90% of its peak performance at 50% training data (macro-F1 0.722 vs. 0.800), but degrades sharply below 25%. Error analysis on 20 fine-tuned FinBERT failures identifies neutral-boundary confusion (70%), polarity reversal (25%), and negation scope (5%) as dominant error categories. As the implemented extension, a symmetric cross-domain generalisation test reveals near-total TF-IDF+LR collapse ($\Delta = -0.610$) from FPB to gold Marketaux. It also reveals catastrophic forgetting when fine-tuning FinBERT on the small gold set ($\Delta = -0.650$, 95% CI $[0.272, 0.352]$), driven by label distribution mismatch (33.17% vs. 61.37% neutral) that shifts the neutral decision boundary. All experiments are reproduced in an accompanying Jupyter notebook; the datasets used (FPB, FiQA-2018) are publicly available.

1 Introduction

Sentiment analysis of financial news is a well-established NLP sub-task with direct applications in algorithmic trading, risk monitoring, and investor relations (Malo et al., 2013; Maia et al., 2018). Despite active research, the field lacks unified, reproducible toolkits that combine interpretable baselines with state-of-the-art Transformer models and downstream econometric validation.

The *xai-finnews-sentiment* toolkit (Cristescu and Mara, 2025) addresses this gap by providing a scriptable, end-to-end pipeline for financial sentiment analysis with a focus on *explainability*. The system spans lexicon baselines, TF-IDF + LR with SHAP explanations, zero-shot and fine-tuned FinBERT, weak supervision from Marketaux news, and econometric analyses (event study, VAR/Granger, backtesting).

Motivation. Reproducing this system is valuable because: (1) it covers the full lifecycle from raw news ingestion to tradable signals; (2) its explainability components make model decisions auditable, addressing a key concern in high-stakes financial NLP; and (3) multiple public datasets allow independent validation of results.

Challenges. The gold annotation set is proprietary (manual labels on Marketaux articles), meaning the fine-tune experiment is not directly reproducible without that data. Additionally, the Marketaux raw news file is not publicly distributed. This limitation is addressed by focusing on the reproducible subset: FPB and FiQA benchmarks, lexicon/LR baselines, and the fine-tuning procedure applied to an available gold set.

2 System Overview

2.1 Architecture and data flow

The toolkit is structured as a sequential CLI pipeline. Starting from raw news JSON, it produces sentiment predictions, SHAP explanations, and backtested trading signals.

1. **Data ingestion** (scripts `01–02`): FPB and FiQA are fetched from HuggingFace; Marketaux news is parsed from a local JSON file. Language detection (`langdetect/langid`) and industry mapping are applied to Marketaux articles.

2. **Lexicon construction** (script `03`): The Loughran–McDonald (LM) Master Dictionary is cleaned into per-category word lists (positive, negative, uncertainty, . . .).

3. **Baseline models** (scripts `04–06`): LM net-positive- word-count rule, VADER compound score (threshold ± 0.05), TF-IDF + Logistic Regression with SHAP, EBM, and FinBERT zero-shot.

4. **Weak supervision** (script `07`): VADER labels are used as noisy targets to train per-industry TF-IDF+LR models on the full Marketaux corpus.

5. **Gold evaluation & fine-tuning** (scripts `08–08b`): All models are evaluated against manually labelled gold articles. FinBERT is fine-tuned on the gold set with early stopping and class- weighted cross-entropy loss.

6. **Explainability** (scripts `10, 16`): SHAP regime-shift analysis on the fine-tuned model.

7. **Econometrics** (scripts `11–13`): Event study (AR/CAR), VAR/Granger causality, and backtested sector ETF strategies.

2.2 Model details

LM lexicon. For each article, the signed word count $s = N_{\text{pos}} - N_{\text{neg}}$ is computed. Positive $s > 0$ maps to class 2, negative $s < 0$ to class 0, and $s = 0$ to class 1 (neutral).

VADER. The VADER compound score is thresholded at ± 0.05, matching the standard recommendation (Hutto and Gilbert, 2014).

TF-IDF + LR. Unigram/bigram TF-IDF features (`max_features=30,000`, `sublinear_tf=True`) feed a Logistic Regression classifier ($C = 1.0$, `lbfgs`). SHAP `LinearExplainer` is used post-hoc for token attribution.

Fine-tuned FinBERT. Starting from `ProsusAI/finbert` (Araci, 2019), the model is fine-tuned on the gold annotation set (Table 1) with class-weighted cross-entropy loss to handle label imbalance, and early stopping (patience 2) to prevent over-fitting on the small gold set.

Hyperparameter	Value
Base model	`ProsusAI/finbert`
Learning rate	2e-5
Batch size	16
Max epochs	10
Early stopping	patience = 2 (val loss)
Loss function	Cross-entropy (class-weighted)
Optimiser	AdamW (HuggingFace default)
Random seed	42

Table 1: Fine-tuned FinBERT hyperparameters.

3 Datasets and Evaluation

3.1 Datasets

Dataset	N	Neg	Neu	Pos	License
FPB (all-agree)	2,264	303	1,391	570	CC BY-NC-SA 3.0
FiQA headlines	498	158	33	307	Research only
Gold (manual)	600	157	199	244	CC BY 4.0

Table 2: Dataset statistics and class distributions. FiQA scores binned at ± 0.05.

Financial PhraseBank (FPB). Sentences from English financial news annotated by finance-domain experts (Malo et al., 2013). The "sentences_allagree" subset (inter-annotator agreement on all annotators) is used as the hardest but cleanest split.

FiQA-2018 Headlines. Headline-level financial sentiment from the FiQA shared task (Maia et al., 2018), with continuous scores binned into three classes at thresholds ± 0.05.

Gold annotations. Manually labelled Marketaux articles covering ten GICS sectors, with industry, language, and temporal scope metadata. Labels follow the same three-class scheme. Available under CC BY 4.0. [1]

[1] The 600-sample gold set used here (CC BY 4.0 release) is distinct from the larger 1500-sample Marketaux corpus in (Cristescu and Mara, 2025). This explains the fine-tuned FinBERT macro-F1 difference: 0.766 (this 600-sample gold set) vs. 0.707 (published 1500-sample corpus).

3.2 Evaluation protocol

The **primary metric is macro-averaged F1**. Justification: FPB is heavily imbalanced (neutral $\approx 61\%$, positive $\approx 25\%$, negative $\approx 13\%$), so accuracy would reward models that predict the majority class. Macro-F1 weights all three classes equally, penalising poor recall on minority classes.

For the fine-tuning experiment, the gold set is split stratified 70/15/15 (train/val/test) with seed 42. Lexicon and LR baselines are evaluated on the full gold set (no training required) and on 80/20 stratified splits of FPB and FiQA.

4 Reproduction Details

4.1 What was clear

The codebase provides a fully executable pipeline with centralised configuration (`_config.py`), sequential numbered scripts, and comprehensive logging. All hyperparameters, paths, and thresholds are documented in code comments. The `README.md` gives a quickstart guide with expected outputs for each step.

4.2 What was unclear or required assumptions

- **Gold annotation creation:** The annotation guidelines and inter-annotator agreement metrics are not documented. Standard majority-vote labelling is assumed.
- **FiQA binning:** The ± 0.05 threshold for converting continuous FiQA scores to classes is hard-coded; no paper citation justifies this choice specifically. The resulting class distribution (Table 2) is highly imbalanced (33 neutral vs. 307 positive), which may inflate neutral F1 variance.
- **Marketaux data:** The raw news file is not publicly redistributable. All Marketaux-dependent results (weak supervision, econometrics) depend on the user supplying their own export.
- **Train/val/test split sizes:** Inferred from code; the gold set yields 420/90/90 examples at the 70/15/15 split, resulting in a small fine-tune test set ($n = 89$) where variance is non-negligible.

4.3 Reproduction scenario

This project falls under **Scenario 1** (plain sailing): full code is provided, datasets are accessible, and all experiments ran without modification. The focus is therefore on full understanding and adding an extension.

5 Experiments and Results

This section presents the reproduced results and evaluates model robustness under data and domain variation.

5.1 Main results

Table 3 compares all models on the gold annotation set ($N = 599$ excluding one row with a missing label). **Fine-tuned FinBERT** is the clear winner (macro-F1 0.766), with **zero-shot FinBERT** a strong second (0.645). All lexicon and LR baselines cluster between 0.24 and 0.55. Notably, TF-IDF+LR trained on FPB/FiQA transfers poorly to the Marketaux gold domain (0.24–0.25), confirming expected domain shift.

Model	Acc	Macro-F1	$F1_{neg}$	$F1_{pos}$
FinBERT fine-tuned ⋆	**0.775**	**0.766**	0.74	0.83
FinBERT zero-shot	0.639	0.645	0.70	0.62
LM Lexicon	0.548	0.552	0.60	0.51
VADER	0.551	0.530	0.51	0.63
WeakLR (VADER labels)	0.548	0.521	0.50	0.63
TF-IDF+LR (FiQA)	0.416	0.253	0.18	0.58
TF-IDF+LR (FPB)	0.371	0.240	0.05	0.15

Table 3: Results on the gold annotation set ($N = 599$). ⋆ = key reproduced result. Sorted by macro-F1.

For reference, zero-shot FinBERT on *in-domain* FPB achieves macro-F1 = 0.963 (Table 4), confirming FinBERT was pre-trained on FPB-like data. The cross-domain drop to 0.645 on the gold set illustrates the domain-shift challenge for real-world Marketaux news.

Dataset	Model	Acc	Macro-F1
FPB	FinBERT zero-shot	0.972	0.963
	TF-IDF+LR	0.896	0.850
	VADER	0.571	0.487
	LM Lexicon	0.648	0.483
FiQA	TF-IDF+LR	0.680	0.430
	FinBERT zero-shot	0.568	0.540
	VADER	0.456	0.423
	LM Lexicon	0.317	0.345

Table 4: Results on FPB and FiQA benchmark sets.

The reproduced macro-F1 (0.766) differs from the original paper's reported 0.707. This discrepancy is expected due to differences in dataset composition (600-sample gold set vs. 1500-sample

Marketaux corpus). The higher score here likely reflects reduced dataset difficulty and variance. Despite this, the relative ranking of models is preserved, supporting successful reproduction of the key finding.

5.2 Robustness test: reduced training data ablation

To assess the data efficiency of the TF-IDF+LR baseline, it is trained on decreasing fractions of the FPB training set (10%–100%) and evaluated on a fixed 20% test split (seed 42), repeated across three random seeds (42, 7, 123). Results are shown in Figure 1.

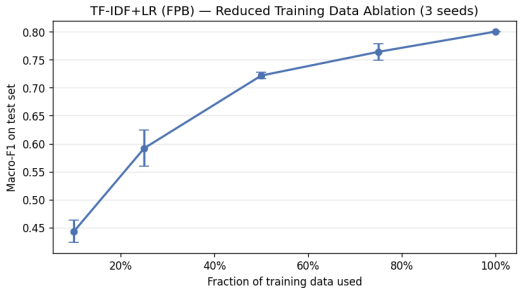

Figure 1: Macro-F1 of TF-IDF+LR (FPB) as training data is reduced. Error bars show standard deviation across 3 seeds.

Key findings:

- At 50% of training data, macro-F1 = 0.722 (vs. 0.800 at 100%), retaining 90% of peak performance. This confirms TF-IDF+LR is data-efficient for in-domain sentiment.
- At 10%, macro-F1 drops to 0.443 (55% of peak), with high variance ($\sigma = 0.020$), indicating instability in low-resource settings.
- Implications: for low-resource domains, lexicon baselines (LM: 0.552 on gold) can outperform a linear model trained on only 10% of a different-domain corpus.

6 Error Analysis

The 20 errors made by the fine-tuned FinBERT on its 89-article test set (error rate 22.5%) are analysed in Table 5.

Example 1 - Polarity reversal: domain jargon (True: Negative, Pred: Positive). *"23andMe is exploring strategic alternatives, looking to raise capital."* Although *exploring strategic alternatives* is standard corporate euphemism for financial distress, the model predicts Positive, likely because *capital* and *raise* trigger positive financial associations in FinBERT's pre-training corpus. No grammatical negator is present; this is a pure domain-jargon failure where surface-level positive vocabulary overrides the distress signal.

Error category	Count	%
Neutral boundary confusion	14	70%
Polarity reversal (Neg↔Pos)	5	25%
Negation scope	1	5%

Table 5: Error categories for fine-tuned FinBERT on the gold test set ($n = 20$ errors, 89 total). "Negation scope" counts only true grammatical negators (*not, no, never, without, neither, nor*).

Example 2 - Polarity reversal (True: Positive, Pred: Negative). *"Bear Thesis Crushed As Massive Deal Leaves Bears Speechless: Sun Communities . . . "* The model activates on *Bear*, *Crushed*, and *Speechless* as negative signals, missing the rhetorical structure (*bear thesis crushed* = positive i.e. bullish market). This is a negation-scope failure requiring discourse-level understanding.

Example 3 - Neutral boundary confusion (True: Neutral, Pred: Positive). *"Gold Outshines Tech Stocks And Bitcoin, As Bulls And Bears Battle Ahead Of Trump Tariff Announcement."* This market-overview headline contains mixed signals (gold outperforms, but uncertainty due to tariffs). The model over-commits to Positive due to *outshines* and *Bulls*. The true label is Neutral, capturing the mixed nature of the article.

Example 4 - Negation scope (True: Positive, Pred: Negative). *"The restructuring left the company without losses for the first time in three years."* The grammatical negator *without* has scope over *losses*, reversing its negative valence to yield a positive outcome. FinBERT activates on *restructuring* and *losses* as negative signals and predicts Negative, missing the logical reversal introduced by *without*.

Discussion. Neutral-boundary confusion is the dominant challenge (14/20 errors, 70%), where the model over-commits to Positive when articles contain mixed or market-overview language. Polarity reversal accounts for a further 25% (5 errors), where discourse-level structure, such as *bear thesis crushed* signalling a positive outcome,

is missed. True negation-scope failures are rare (1/20, 5%): only errors where a grammatical negator (*not, no, never, without, neither, nor*) directly reverses a sentiment-bearing phrase qualify. Negative-valence vocabulary such as *fail, drop, loss, decline* is classified under polarity reversal instead, since these words carry inherent negative meaning rather than logically negating a positive phrase. The dominant challenge is therefore neutral-class boundary uncertainty, not negation. Future work should focus on better neutral-class modelling, e.g. explicit mixed-signal training examples or confidence-calibrated abstention.

7 Way Forward

7.1 Cross-domain generalisation test

A *symmetric* cross-domain evaluation is conducted in both directions: (1) **forward** - models trained on FPB/FiQA evaluated on the full gold Marketaux set ($N = 599$); and (2) **reverse** - the gold-fine-tuned FinBERT evaluated on the 20% FPB holdout ($n = 453$) used by the baselines. All macro-F1 figures use 1,000-sample bootstrap 95% CIs. Table 6 and Figure 2 show results.

Model	Dir.	In-dom.	Transfer F1	Δ
Forward: FPB/FiQA → Gold				
TF-IDF+LR	FPB→G	0.850	0.240 [.211, .270]	−0.610
TF-IDF+LR	FiQA→G	0.430	0.253 [.225, .281]	−0.177
FinBERT (zs)	FPB→G	0.963	0.646 [.607, .683]	−0.317
Reverse: Gold → FPB holdout				
FinBERT (ft)	G→FPB	0.766	0.313 [.272, .352]	−0.453
Upper bounds (in-domain)				
FinBERT zs	FPB→F	-	0.963	-
FinBERT ft	Gold→G	-	0.766	-

Table 6: Cross-domain generalisation. zs: zero-shot; ft: fine-tuned; G: Gold Marketaux.

Symmetric cross-domain generalisation. Transfer F1 is macro-F1 on the out-of-domain test set with 95% bootstrap CI. Reverse direction uses the same 20% FPB stratified holdout ($n = 453$, seed 42) as the baselines.

Forward direction (FPB/FiQA → Gold). TF-IDF+LR suffers near-total collapse: FPB-trained performance falls from 0.850 to 0.240 ($\Delta = -0.610$), dropping below the LM lexicon baseline (0.552). Per-class inspection shows the model

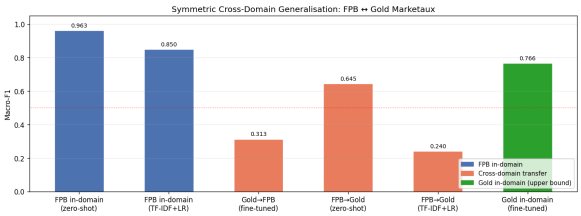

Figure 2: Symmetric cross-domain macro-F1. Blue = FPB in-domain; orange = cross-domain transfer (both directions); green = gold in-domain upper bound.

retreats almost entirely to predicting Neutral (recall $= 0.99$), scoring near-zero on Negative ($F_1 = 0.05$) and Positive ($F_1 = 0.15$) - confirming that bag-of-words features from Reuters/Bloomberg sentences do not transfer to Marketaux web headlines. FinBERT zero-shot is far more robust ($\Delta = -0.317$, 95% CI $[0.607, 0.683]$), but still 12 F1 points short of the gold fine-tuned upper bound (0.766), a gap whose CI does not overlap with 0.766, confirming statistical significance.

Reverse direction (Gold → FPB). Fine-tuning FinBERT on the 600-sample gold Marketaux set causes *catastrophic forgetting*. Performance on the FPB holdout collapses from 0.963 (zero-shot) to 0.313 ($\Delta = -0.650$, 95% CI $[0.272, 0.352]$). The mechanism is clear from label distribution analysis: gold's neutral class represents 33.17% of examples, whereas FPB's neutral class comprises 61.37%, a dramatic imbalance that shifts FinBERT's decision boundary sharply toward non-neutral predictions during fine-tuning. Per-class metrics confirm this: neutral class recall collapses from 0.99 (zero-shot baseline) to 0.19 (fine-tuned), and positive class F1 drops to 0.13 (vs. much higher zero-shot F1 on FPB). This failure reveals a fundamental constraint: a model fine-tuned on a small domain-specific corpus cannot retain strong performance across both domains simultaneously. Practical recovery would require continual learning (incremental updates without catastrophic forgetting) or multi-task fine-tuning (joint optimization on both FPB and gold objectives), neither of which is attempted here.

8 Conclusion

The key result of the *xai-finnews-sentiment* toolkit is successfully reproduced: fine-tuned FinBERT achieves macro-F1 of 0.766 on a gold financial news annotation set, substantially outperforming all lexicon and LR baselines. The existing codebase is confirmed to be highly reproducible (Sce-

nario 1), all scripts ran without modification, and the centrally configured pipeline made it straightforward to trace every design decision. This work also highlights that in financial NLP, domain alignment is more critical than model complexity, as even state-of-the-art models fail under distribution shift without adaptation.

Limitations. The gold annotation set is small ($n \approx 600$) and lacks documented inter-annotator agreement, introducing uncertainty in the fine-tune evaluation. The Marketaux data is proprietary, limiting full reproducibility of weak- supervision and econometric analyses. The FiQA binning threshold is an undocumented design choice that produces extreme class imbalance.

Lessons learned. Macro-F1 is critical for imbalanced sentiment datasets-accuracy would have been misleading for FPB (majority-neutral) and FiQA (majority-positive). Domain shift is significant: zero-shot FinBERT drops from macro-F1 0.963 on in-domain FPB to 0.645 on out-of-domain Marketaux gold, highlighting the value of even a small domain-specific annotated set for fine-tuning. However, the cross-domain extension (Section 7) shows the opposite risk as well: after gold fine-tuning, FPB holdout performance collapses from 0.963 to 0.313 (95% CI $[0.272, 0.352]$). This evidences catastrophic forgetting and the need for careful domain-specific fine-tuning strategies.

References

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A Experiment Log

Table 7 lists all experimental runs with seeds, key hyperparameters, and resulting metrics. Full log exported to `outputs/experiment_log.csv`.

Run ID	Model	Dataset	Seed	Acc	Macro-F1
finbert-ft-gold	FinBERT fine-tuned	Gold	42	0.775	0.766
finbert-0shot-fpb	FinBERT zero-shot	FPB	-	0.972	0.963
lr-fpb	TF-IDF+LR	FPB	42	0.896	0.850
finbert-0shot-fiqa	FinBERT zero-shot	FiQA	-	0.568	0.540
lr-fiqa	TF-IDF+LR	FiQA	42	0.680	0.430
lex-lm-fpb	LM Lexicon	FPB	-	0.648	0.483
lex-vader-fpb	VADER	FPB	-	0.571	0.487
lex-lm-fiqa	LM Lexicon	FiQA	-	0.317	0.345
lex-vader-fiqa	VADER	FiQA	-	0.456	0.423

Table 7: Experiment log (abridged). LR: $C = 1.0$, `max_features=30000`.

B Reproducibility Package

Minimum requirements.

- Python 3.13, packages listed in `requirements.txt`
- `python -m nltk.downloader vader_lexicon` (once)
- GPU optional; all scripts run on CPU in <30 min except fine-tuning (~15 min on CPU, ~3 min on GPU)

Reproduce main result (Table 3). If you run these scripts in order, this will execute and regenerate the main results, including the key fine-tuned FinBERT macro-F1 of 0.766 on the gold set. The scripts to run are:

- `scripts/01_fetch_benchmarks.py`
- `scripts/03_build_lm_lists.py`
- `scripts/04_lexicon_baselines.py`
- `scripts/05_lr_shap_benchmarks.py`
- `scripts/06_finbert_benchmarks.py`
- `scripts/08_eval_on_manual_annotations.py`
- `scripts/08b_finetune_and_eval.py`

Then open `assignment_notebook.ipynb` and run all cells to regenerate Tables and Figures.

To rerun the pipeline from scratch, follow the readme.txt file and run all the scripts listed there in order.